Student Management System – Algorithm

❖ Main Algorithm

Step 1: Start
Step 2: Initialize:

- Student array
- Stack (for undo)
- Queue (for reports)
- studentCount = 0

Step 3: Repeat
  Step 3.1: Display menu
  Step 3.2: Input user choice

Step 3.3: Perform operation based on choice:
  1 → Insert Student
  2 → View All Students
  3 → Search by ID
  4 → Search by Name
  5 → Delete Student
  6 → Update Student
  7 → Sort Students
  8 → Display Statistics
  9 → Undo Last Action
  10 → Exit

Step 4: Continue until choice = 10
Step 5: End

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❖ *Insert Student Algorithm*

Step 1: Start
Step 2: Check if student list is full
  → If full, display message and stop
Step 3: Input student ID
Step 4: Check ID uniqueness
  → If not unique, re-enter ID
Step 5: Input name
Step 6: Input age
  → Validate (1–100), else re-enter

Step 7: Input course
Step 8: Store student in array
Step 9: Push ("insert") to stack
Step 10: Increment studentCount
Step 11: Display success message
Step 12: End

- ***View All Students Algorithm***

Step 1: Start
Step 2: If no students → Display message
Step 3: Else display all student records
Step 4: End

- ***Search Student by ID***

Step 1: Start
Step 2: Input student ID
Step 3: Traverse student list
Step 4: If found → Display details
Step 5: Else → Display "Not Found"
Step 6: End

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- ***Search Student by Name***

Step 1: Start
Step 2: Input name
Step 3: Traverse student list
Step 4: If match → Display details
Step 5: Else → Display "Not Found"
Step 6: End

- ***Delete Student Algorithm***

Step 1: Start
Step 2: Input student ID
Step 3: Search student
Step 4: If found:
  a. Push ("delete") to stack
  b. Shift elements left

c. Decrease studentCount
d. Display success message

Step 5: Else → Display "Not Found"
Step 6: End

❖ *Update Student Algorithm*

Step 1: Start
Step 2: Input student ID
Step 3: Search student
Step 4: If found:
a. Push old data to stack
b. Input new name
c. Input new age
d. Input new course
e. Update record

Step 5: Else → Display "Not Found"
Step 6: End

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❖ *Undo Last Action Algorithm*

Step 1: Start
Step 2: Check if stack is empty
→ If yes, display "Nothing to undo"
Step 3: Pop last action
Step 4: If action = "insert"
→ Remove last student
Step 5: If action = "delete"
→ Restore deleted student
Step 6: Display success message
Step 7: End

❖ *Sort Students Algorithm (Bubble Sort)*

Step 1: Start
Step 2: Repeat for all students
Step 3: Compare adjacent names
Step 4: If out of order → Swap
Step 5: Continue until sorted

Step 6: Display success message
Step 7: End

- ***Display Statistics Algorithm***

Step 1: Start
Step 2: Enqueue "Statistics"
Step 3: While queue not empty:
    a. Dequeue request
    b. If no students → Display message
    c. Calculate total age
    d. Compute average age
    e. Display results
Step 4: End

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